

Tane Koh

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Education

Olin College of Engineering

May 2027

B.S. in Electrical and Computing Engineering

Relevant Coursework: Microelectronic Circuits, Principles of Integrated Engineering, Sensors/Instrumentation, Discrete Math, Computer Architecture, Engineering Systems Analysis: Signals

Skills

Programming: MATLAB, SystemVerilog, Python, JavaScript, HTML, R, C++, Arduino, RasPi, MySQL

Design: SolidWorks, KiCad, Adobe Creative Cloud, Microsoft Suite, LaTeX, LTSpice

Fabrication: Bandsaw, Drill press, Miter Saw, 3D Printer, Jig Saw, Table Saw, Laser Cutter/Printer, Screen Printer, Sublimation Printer, Knitting

Experience

Principles of Integrated Engineering: Vending Machine Project

Sep 2024 - Dec 2024

Electrical Engineer

- Prototyped and drafted a circuit to power stepper motors with motor drivers and a current-mode converter
- Constructed circuitry to control and power a lighting system, stepper motors, and proximity sensors
- Integrated and programmed a microcontroller and microprocessor to control touchscreens and process signal input

Differential Amplifier Simulation and Measurement Project

Jan 2025 - May 2025

Electrical Engineer

- Designed and analyzed a Class-AB transconductor-based differential amplifier to improve slew rate performance
- Simulated and validated circuit behavior and characteristics in LTSpice, including Delbruck's bump/anti-bump circuit
- Implemented a folded-cascode/flipped-follower configuration to maintain constant bump current

Multicycle Unpipelined RISC-V Processor

Sep 2025 - Dec 2025

Engineer and Project Manager

- Designed a fully functional RISC-V processor in System Verilog capable of supporting all instruction types
- Developed and decoded modular components, including a 32-bit register file and a 32-bit ALU with flag outputs
- Verified capability through simulation using Icarus Verilog, demonstrating proper state transitions and operations

COUCH Lab Personal Project

July 2025 - Present

Electrical Engineer

- Designed and refined circuitry to power two 1800W bi-directional DC motors safely
- Reconfigured a PS4 controller to control throttle through an ESP32 board, mapping DAC pins to motor controls
- Integrated circuitry with software, using Hall sensor readings to implement PID control

Discrete Math: Combinatorics and Game Theory

Sep 2025 - Dec 2025

Teaching Assistant

- Tutored students on advanced counting techniques, induction, recurrence relations, and discrete structures such as graphs and trees
- Graded homework, pre-class assignments, and final projects

Introduction to Sensors, Instrumentation, and Measurement

Jan 2025 - May 2025

Teaching Assistant

- Codesigned problem sets and labs on electrical measurements, RC filters, op-amp circuits, and measurement systems
- Led lectures and labs, guiding students through sensor interfacing, circuit troubleshooting, and data collection
- Aided student teams in designing capstone experiments integrating sensors, signal conditioning, and data analysis

Olin College As a Living Lab: Sustainability Committee

Jan 2024 - May 2025

Student Intern

- Conducted research evaluating opportunities for upgrading operations with sustainable infrastructure at Olin
- Contacted partnered companies to implement upgrades alongside students as curricular projects
- Collaborated with staff in developing Olin's strategic plan to enact sustainability initiatives across campus